

Student API — Full Project Documentation

Project: Student CRUD REST API (SOC Practical Exam)

Local folder: C:\Users\rasar\OneDrive\Desktop\SOC

GitHub user: [Razaara](#)

GitHub repo: <https://github.com/Razaara/student-api>

Docker Hub user: razzara

Docker image: razzara/student-api:latest

Date built & verified: 10 August 2026

1. What was built

A complete A→Z Student CRUD REST API matching the exam handbook:

Layer	File
Main	StudentApiApplication.java
Entity	entity/Student.java
Repository	repository/StudentRepository.java
Service	service/StudentService.java + StudentServiceImpl.java
Controller	controller/StudentController.java
Config	application.properties (port 8500)
Docker	Dockerfile (multi-stage, Java 21)
Compose	docker-compose.yml (MySQL 8 + Spring Boot)
CI	Jenkinsfile (Declarative pipeline)

Endpoints

Method	Path	Success code	Tested
POST	/api/students	201 Created	Yes
GET	/api/students	200 OK	Yes
GET	/api/students/{id}	200 OK	Yes
PUT	/api/students/{id}	200 OK	Yes
DELETE	/api/students/{id}	204 No Content	Yes

Missing id currently returns **500** (unhandled `RuntimeException`) — same as the handbook sample code.

2. Step-by-step work log (everything done)

Step A — Workspace setup

1. Used existing folder Desktop\SOC (already had .git).
2. Moved Cursor agent root into SOC .
3. Confirmed tools: Java 21, Maven 3.9.16, Docker 29.x, Git.

Step B — Scaffold Spring Boot project

1. Tried Spring Initializr download → **failed** (400 Bad Request).
2. **Fix recorded:** start.spring.io currently only offers Spring Boot **4.x**. Handbook targets Boot **3.x**, so the project was scaffolded manually with:

```
<parent>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-parent</artifactId>
  <version>3.4.5</version>
</parent>
```

3. Created packages:
 - com.example.studentapi
 - ...entity / ...repository / ...service / ...controller
4. Added dependencies: Web, Data JPA, MySQL connector (mysql-connector-j), Lombok, Validation.
5. Wrote all source files from the corrected handbook.

Step C — Database

1. Local Windows service MySQL97 was **Stopped**.
2. **Fix recorded:** started MySQL via Docker Compose instead of the local service:

```
docker compose up -d mysql
```

3. Set credentials for Docker MySQL:
 - username: root
 - password: root
4. **Fix recorded vs handbook empty password:** handbook showed spring.datasource.password= (empty). Empty password fails against Docker MySQL (MYSQL_ROOT_PASSWORD=root). Updated application.properties to password=root and documented it.

Step D — Local Maven build & run

```
mvn clean package -DskipTests # SUCCESS
mvn spring-boot:run           # Tomcat on 8500
```

Step E — Local CRUD testing (verified)

Test	Result
POST AS2023000 John Doe	201 + JSON body
POST AS2023001 Jane Smith	201 + JSON body
GET all	200 with both students

GET by id AS2023000	200
PUT update John → John Updated	200, id stayed AS2023000
DELETE AS2023001	204
GET missing id	500 (expected for sample code)

Fix recorded (test tooling): PowerShell `curl.exe -d "{\"id\":\"...\"}` mangled JSON and produced HTTP 400. Retested successfully with `Invoke-RestMethod` and proper JSON strings.

Step F — Docker image build

1. First `docker compose build` **failed** on:

```
mvn dependency:go-offline
Premature end of Content-Length delimited message body
```

2. **Fix recorded:** simplified `Dockerfile` — removed fragile `dependency:go-offline` step; copy sources and run a single `mvn -B clean package -DskipTests` .
3. Rebuild succeeded → image `soc-springboot:latest` .

Step G — Docker Compose full stack testing

```
docker compose up -d --build
```

Containers:

Name	Image	Ports	Status
student-mysql	mysql:8.0	3306:3306	healthy
student-api	soc-springboot	8500:8500	running

Docker CRUD retested:

Test	Result
POST DCK001	201
GET all	200 (includes prior local data via MySQL volume)
PUT DCK001	200
DELETE DCK001	204

Step H — GitHub repository

1. Created public repo via GitHub API as user **Razaara**.
2. Repo URL: <https://github.com/Razaara/student-api>
3. Cleaned a stale gitlink entry named `studentapi` left from an earlier incomplete init (**fix recorded**).
4. Committed and pushed `main` :

Add complete Student CRUD Spring Boot project for SOC practical exam.

Step I — Docker Hub push

```
docker tag soc-springboot:latest razzara/student-api:latest
docker push razzara/student-api:latest
```

Image: `docker.io/razzara/student-api:latest`

Step J — Documentation

- `README.md` in the repo (quick start).
 - This `PROJECT_DOCUMENTATION.md` (full A→Z log + fixes).
 - PDF generated from this file for printing / exam reference.
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3. How to run (exam-ready)

Option 1 — Local app + Docker MySQL

```
cd C:\Users\rasar\OneDrive\Desktop\SOC
docker compose up -d mysql
mvn spring-boot:run
```

Open Postman: `http://localhost:8500/api/students`

Option 2 — Full Docker Compose

```
cd C:\Users\rasar\OneDrive\Desktop\SOC
docker compose up -d --build
```

Option 3 — Pull from Docker Hub

```
docker pull razzara/student-api:latest
# still need a MySQL instance; prefer Option 2 for exam
```

Sample Postman body (POST)

```
{
  "id": "AS2023000",
  "name": "John Doe",
  "email": "john@example.com",
  "age": 21
}
```

4. application.properties (final working)

```
spring.application.name=studentapi
spring.datasource.url=jdbc:mysql://localhost:3306/students?createDatabaseIfNotExist=true
spring.datasource.username=root
spring.datasource.password=root
spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update
server.port=8500
```

Inside Compose, Spring overrides datasource via env vars to use host `mysql` (service name), not `localhost` .

5. Dockerfile (final working)

```
FROM maven:3.9-eclipse-temurin-21 AS build
WORKDIR /app
COPY pom.xml .
COPY src ./src
RUN mvn -B clean package -DskipTests

FROM eclipse-temurin:21-jre-alpine
WORKDIR /app
COPY --from=build /app/target/*.jar app.jar
EXPOSE 8500
ENTRYPOINT ["java", "-jar", "app.jar"]
```

6. docker-compose.yml (final working)

- Service `mysql` with healthcheck
 - Service `springboot` depends on healthy MySQL
 - Port map `8500:8500`
 - Volume `mysql_data` for persistence
 - Datasource URL uses `jdbc:mysql://mysql:3306/students?...`
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7. Jenkinsfile notes

Declarative pipeline stages:

1. Checkout (`https://github.com/Razaara/student-api.git`)
2. Maven package
3. Docker build
4. Stop/remove old container (`|| exit 0` for Windows)
5. Run new container on `8500:8500`

Requires Jenkins tools named `maven3` and `jdk21` .

8. Fixes summary (must remember for exam)

#	Issue	Fix applied
1	Spring Initializr only offers Boot 4	Manual Boot 3.4.5 project
2	Local MySQL service stopped	Use Docker MySQL (<code>docker compose up -d mysql</code>)
3	Empty DB password in handbook	Set password to <code>root</code> for Docker MySQL
4	Port confusion 8080 vs 8500	Standardized on 8500 everywhere
5	Dockerfile <code>go-offline</code> network flake	Single <code>mvn package</code> step
6	Compose YAML incomplete in old handbook	Full MySQL + Spring services with healthcheck
7	PowerShell JSON escaping broke POST tests	Use <code>Invoke-RestMethod</code> / correct JSON strings
8	Stale gitlink <code>studentapi</code> in repo	Removed with <code>git rm --cached</code> before commit
9	Docker Hub username	Used razzara (not the GitHub username)
10	GitHub username	Used Razaara for the new repo

9. Verification checklist

- ☒ Maven package succeeds
- ☒ App starts on port 8500
- ☒ POST / GET / PUT / DELETE work locally
- ☒ Docker image builds
- ☒ Docker Compose stack healthy
- ☒ CRUD works against Compose stack
- ☒ GitHub repo created: <https://github.com/Razaara/student-api>
- ☒ Code pushed to `main`
- ☒ Image tagged/pushed: `razzara/student-api:latest`
- ☒ Documentation + PDF produced

10. Useful commands cheat sheet

```
# Git
git add .
git commit -m "message"
git push -u origin main

# Maven
mvn spring-boot:run
mvn clean package -DskipTests

# Docker
docker compose up -d --build
```

```
docker compose ps
docker compose logs -f springboot
docker compose down

docker tag soc-springboot:latest razzara/student-api:latest
docker push razzara/student-api:latest
docker pull razzara/student-api:latest
```

11. Links

- GitHub: <https://github.com/Razaara/student-api>
- Docker Hub: <https://hub.docker.com/r/razzara/student-api>
- Local project: C:\Users\rasar\OneDrive\Desktop\SOC